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3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture Week 3: Data Warehouse and Online Analytical Processing

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Contents

- Data Warehouse: Basic Concepts
- Data Warehouse Modeling: Data Cube and OLAP
- Data Warehouse Implementation
- Data Generalization by Attribute-Oriented Induction
- Summary

What is a Data Warehouse?

- Defined in many different ways, but not rigorously.
  - A decision support database that is maintained separately from the organization's operational database
  - Support information processing by providing a solid platform of consolidated, historical data for analysis.
- "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—William H. Inmon *
- Data warehousing:
  - The process of constructing and using data warehouses

*An American computer scientist, recognized by many as the **father of the data warehouse**

Data Warehouse—Subject-Oriented

- Organizes around major subjects, such as customer, product, sales
- Focuses on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provides a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process
  - E.g. “sales” can be a particular subject to be analyzed

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources (*multiple data sources*), such as
  - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
    - E.g. source A and source B may have different ways of identifying a product, but in a data warehouse, there will be only a single way of identifying a product..
  - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the *data warehouse* is significantly longer than that of *operational systems*
  - Operational database: current or most recent value data
  - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
  - Contains an element of time, explicitly or implicitly
  - But the key of operational data may or may not contain "time element"

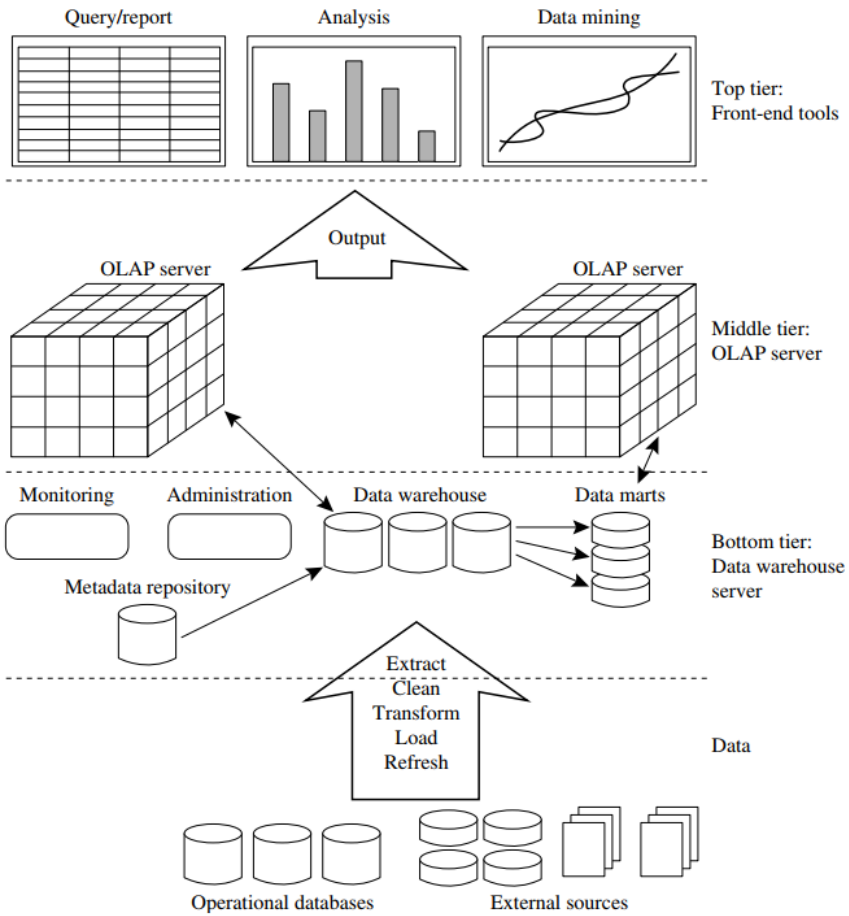
Data Warehouse—Nonvolatile

- A physically separate store of data transformed from the operational environment
- Operational update of data does not occur in the data warehouse environment
  - Historical data in a data warehouse should never be altered
  - Does not require transaction processing, recovery, and concurrency control mechanisms
  - Requires only two operations in data accessing:
    - *initial loading of data* and *access of data*

Why a Separate Data Warehouse?

- High performance for both systems (operational vs warehouse)
  - DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
  - Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- Different functions and different data:
  - missing data: Decision support requires historical data which operational DBs do not typically maintain
  - data consolidation: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources which operational DBs have raw data
  - data quality: Different sources typically use inconsistent data representations, codes and formats which have to be reconciled

Data Warehouse: A Multi-Tiered Architecture



The top tier is a front-end client layer, which contains query and reporting tools, analysis tools, and/or data mining tools (e.g., trend analysis, prediction, visualization, and so on).

The middle tier is an OLAP server that is typically implemented using either (1) a relational OLAP (ROLAP) model; or (2) a multidimensional OLAP (MOLAP) model.

The bottom tier is a warehouse database server that is almost always a relational database system. This tier also contains a metadata repository, which stores information about the data warehouse and its contents.

Back-end tools and utilities perform data extraction, cleaning, and transformation (e.g., to merge similar data from different sources into a unified format), as well as load and refresh functions to update the data warehouse. (ETL process)

Extraction, Transformation, and Loading (ETL)

- **Data extraction**
  - get data from multiple, heterogeneous, and external sources
- **Data cleaning (process missing, inconsistent, noisy, ….)**
  - detect errors in the data and rectify them when possible
- **Data transformation**
  - convert data from legacy or host format to warehouse format
- **Load**
  - sort, summarize, consolidate, compute views, check integrity, and build index and partitions
- **Refresh**
  - propagate the updates from the data sources to the warehouse

Three Data Warehouse Models

From the architecture point of view, there are three data warehouse models:

- Enterprise warehouse
  - collects all of the information about subjects spanning the entire organization
  - provides corporate-wide data integration, usually from one or more operational systems or external information providers
  - requires extensive business modeling and may take years to design and build.
- Data Mart
  - a subset of corporate-wide data that is of value to a specific groups of users.
  - Its scope is confined to specific, selected groups, such as marketing data mart
  - The implementation cycle is more likely to be measured in weeks.
  - Depending on the source of data, data marts can be categorized as
    - **Independent** (from one or more operational systems or external information providers)
    - **Dependent** (directly from warehouse) data mart
- Virtual warehouse
  - A set of views over operational databases
  - Only some of the possible summary views may be materialized

Data Warehousing and On-line Analytical Processing

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Multidimensional Data Model

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- The multidimensional data model
  - an integral part of OLAP.
  - designed to solve complex queries in real time.
  - allows users to understand databases easily
  - facilitates applications to efficiently access databases
  - Contains:
    - Data cubes (data structure)
    - Measures
    - Dimensions
    - Hierarchies
    - Levels

From Tables and Spreadsheets to Data Cubes

- A **data warehouse** is based on a multidimensional data model which views data in the form of a data cube
- A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions, e.g. time, item, branch, and location (buy bread, morning, Coles).
  - **Dimension table**
    - further describes dimension, such as item (item_name, brand, type), or time (day, week, month, quarter, year) or customer (cust_id, cust_firstname, cust_lastname, contact)
  - **Fact table** contains **measures** (such as dollars_sold) and keys to each of the related dimension tables such as sale_records (item, time, cust_id, product_id, quantity)

From Tables and Spreadsheets to Data Cubes

Table 4.2 2-D View of Sales Data for *AllElectronics* According to *time* and *item*

	location = *"Vancouver"*			
	item *(type)*			
time *(quarter)*	*home entertainment*	*computer*	*phone*	*security*
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580

Note: The sales are from branches located in the city of Vancouver. The measure displayed is *dollars_sold* (in thousands).

- In this 2-D representation, the sales for Vancouver are shown with respect to the time dimension (organized in quarters) and the item dimension (organized according to the types of items sold). The fact or measure displayed is dollars sold (in thousands)

From Tables and Spreadsheets to Data Cubes

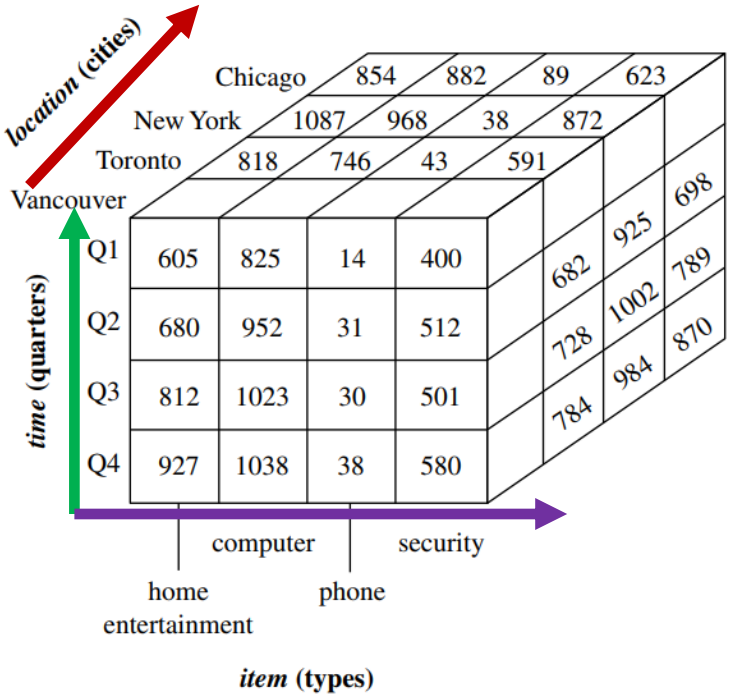
Table 4.3 3-D View of Sales Data for *AllElectronics* According to *time*, *item*, and *location*

	location = "Chicago"				***location*** = "New York"				***location*** = "Toronto"				***location*** = "Vancouver"			
	item				***item***				***item***				***item***			
	home				*home*				*home*				*home*			
time	*ent.*	*comp.*	*phone*	*sec.*	*ent.*	*comp.*	*phone*	*sec.*	*ent.*	*comp.*	*phone*	*sec.*	*ent.*	*comp.*	*phone*	*sec.*
Q1	854	882	89	623	1087	968	38	872	818	746	43	591	605	825	14	400
Q2	943	890	64	698	1130	1024	41	925	894	769	52	682	680	952	31	512
Q3	1032	924	59	789	1034	1048	45	1002	940	795	58	728	812	1023	30	501
Q4	1129	992	63	870	1142	1091	54	984	978	864	59	784	927	1038	38	580

Note: The measure displayed is *dollars_sold* (in thousands).

- Now, suppose that we would like to view the sales data with a third dimension. For instance, suppose we would like to view the data, according to *time* and *item*, as well as *location*, for the cities Chicago, New York, Toronto, and Vancouver. The 3-D data in the table are represented as a series of 2-D tables. Conceptually, we may also represent the same data in the form of a 3-D data cube.

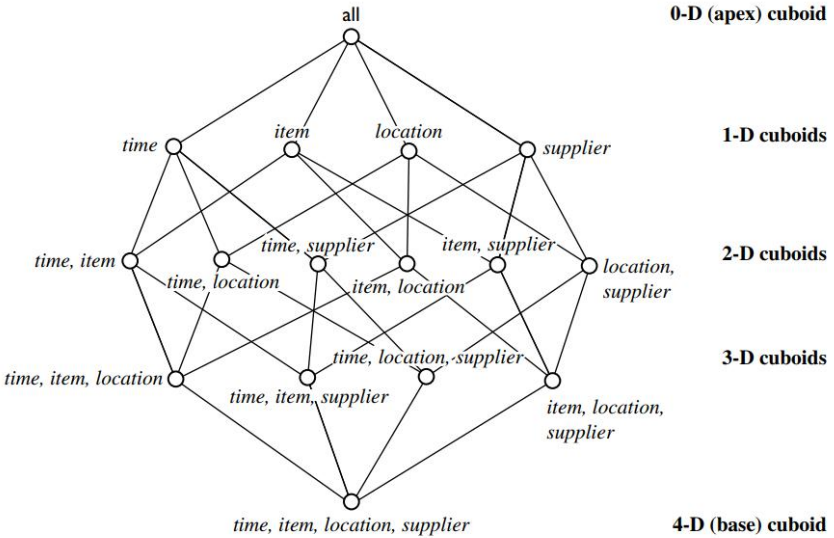
From Tables and Spreadsheets to Data Cubes



A 3-D data cube representation of the data in Table 4.3, according to *time*, *item*, and *location*. The measure displayed is *dollars_sold* (in thousands).

Cube: A Lattice of Cuboids

- Given a set of dimensions, we can generate a cuboid for each of the possible subsets of the given dimensions. The result would form a lattice of cuboids, each showing the data at a different level of summarization, or group-by. The lattice of cuboids is then referred to as a data cube.



Lattice of cuboids, making up a 4-D data cube for *time*, *item*, *location*, and *supplier*. Each cuboid represents a different degree of summarization.

The 0-D cuboid, which holds the highest level of summarization, is called the **apex cuboid**. In our example, this is the total sales, or *dollars sold*, summarized over all four dimensions. The apex cuboid is typically denoted by **all**.

The cuboid that holds the lowest level of summarization is called the **base cuboid**.

This example shows a lattice of cuboids forming a data cube for the dimensions time, item, location, and supplier.

Conceptual Modeling of Data Warehouses

- **Three forms of multidimensional model** in data warehouse
  - Star schema: A fact table in the middle connected to a set of dimension tables
  - Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
  - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

Measures

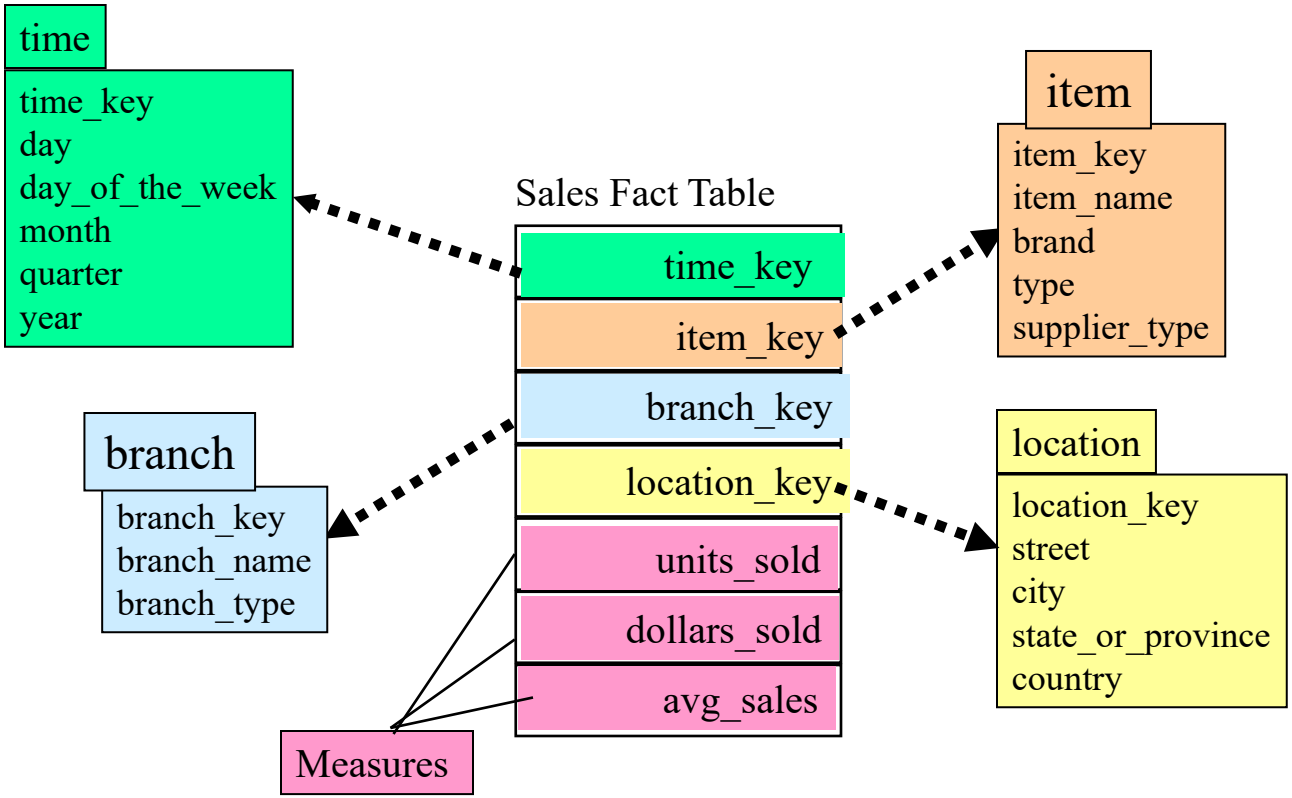
- A measure is an attribute on which aggregate calculations (sum, max, min, mean, etc.) can be made.

Car_sales table

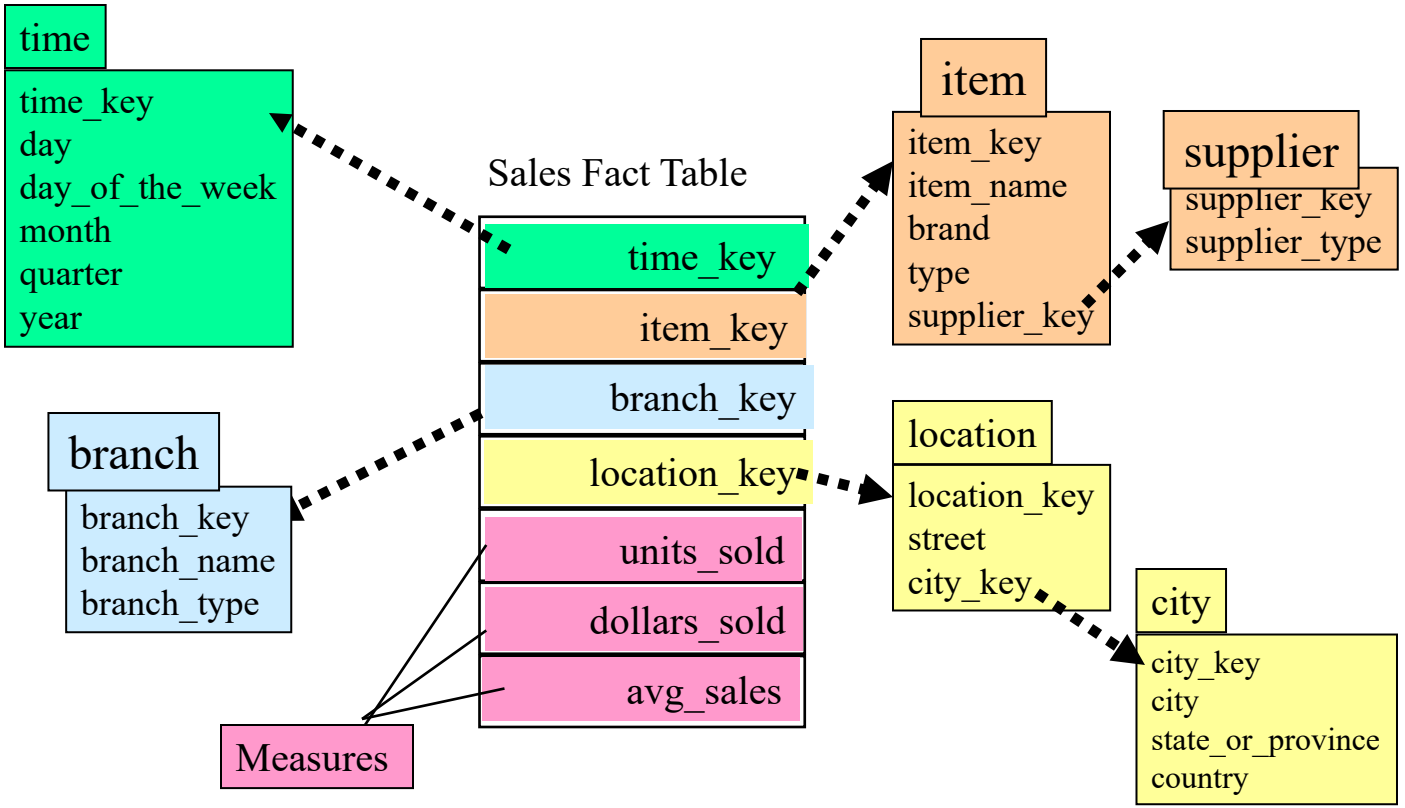
Receipt_no	make	color	size	date	price
0001	Toyota	white	small	01/01/01	20000
0002	Nissan	red	Medium	01/01/01	25000
0003	Holden	silver	Medium	01/02/01	299000
….	…..	…..	…..	…..	……

Is “Date” a measure?

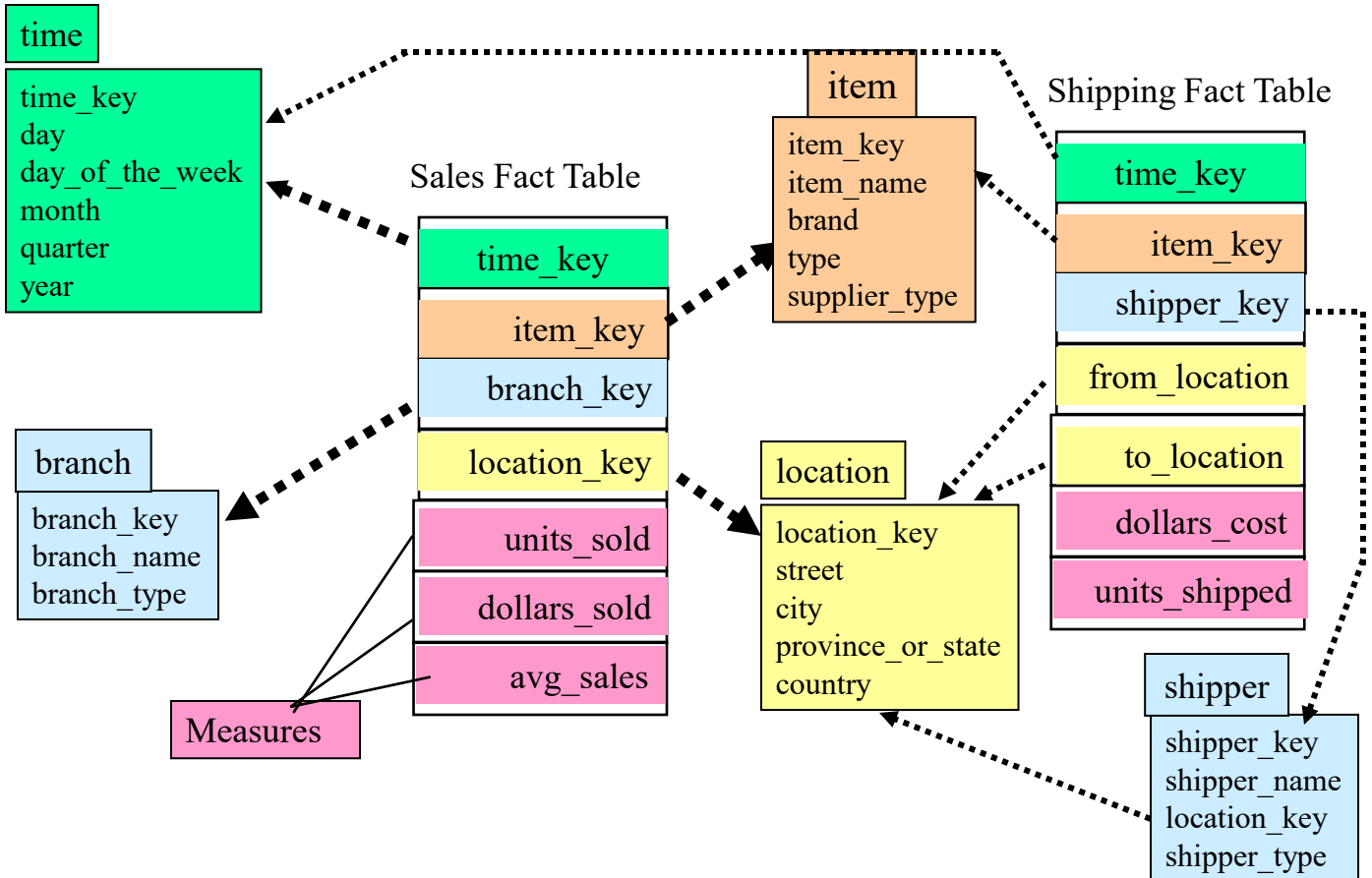
Example of **Star Schema**



Example of **Snowflake Schema**

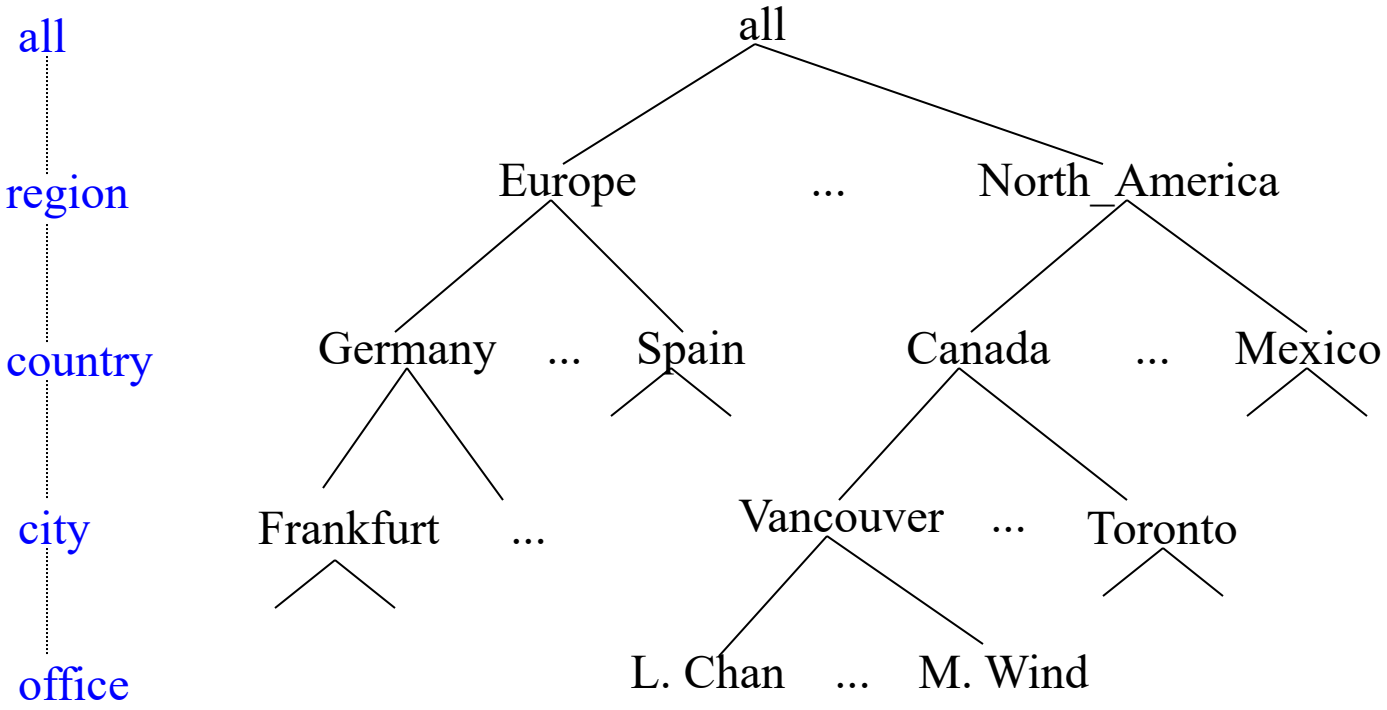


Example of **Fact Constellation**



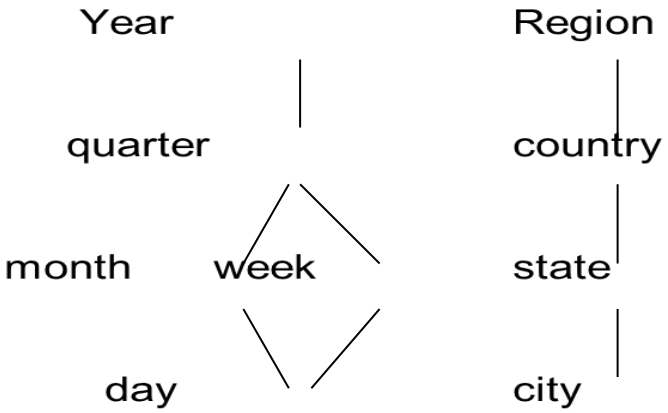
A Concept Hierarchy: **Dimension** (Location)

- A concept hierarchy defines a sequence of mappings, from a set of low-level concepts to higher-level, more general concepts.



A Concept Hierarchy: **Dimension** (Time & Location)

- A **hierarchy** is a way to organize data at different levels of aggregation.
- **A level** represents a position in the hierarchy.
  - For each dimension, the set of values can be organized in a hierarchy.
    - E.g., assuming the sales table includes time and location dimension



Base level (most detailed)

- The hierarchies represent different levels of details for attributes.

Data Cube Measures: Three Categories

"How are measure values computed?" -- by measure functions.

A data cube measure is a numeric function that can be evaluated at each point in the data cube space. The types of measure can be:

- Distributive: if the result derived by applying the function to *n* aggregate values is the same as that derived by applying the function on all the data without partitioning
  - E.g., count(), sum(), min(), max()
- Algebraic: if it can be computed by an algebraic function with *M* arguments (where *M* is a bounded integer), each of which is obtained by applying a distributive aggregate function
  - E.g., avg(), min_N(), standard_deviation()
- Holistic: if there is no constant bound on the storage size needed to describe a subaggregate.
  - E.g., median(), mode(), rank()

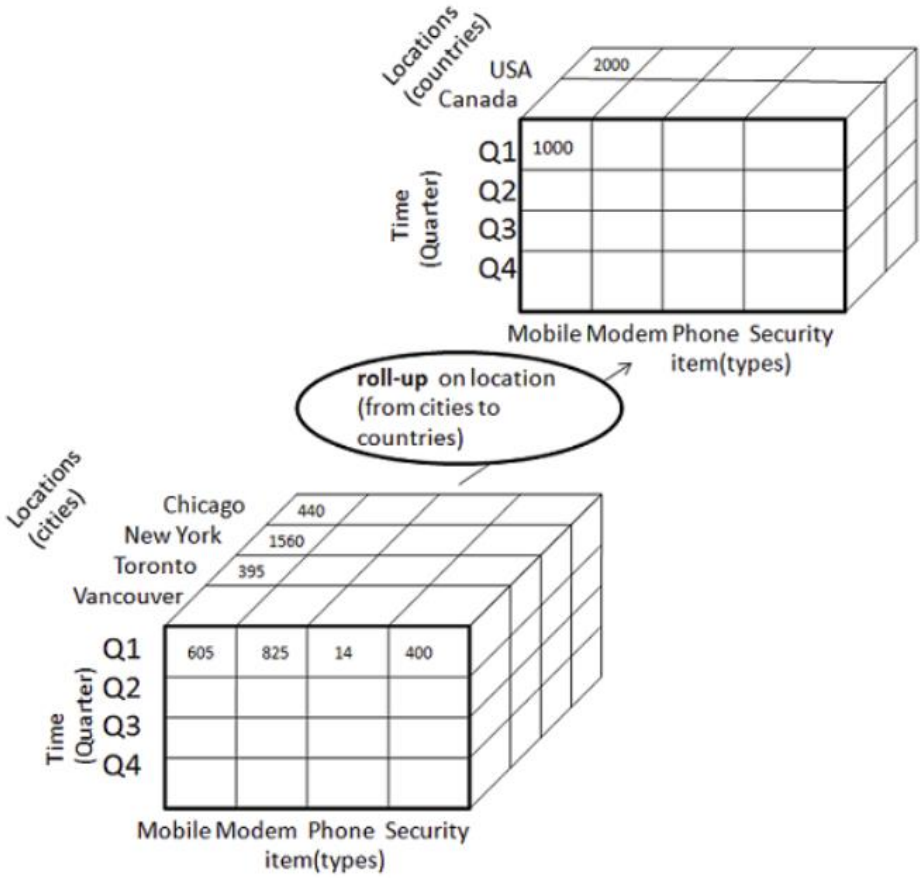
Typical OLAP Operations

"How are concept hierarchies useful in OLAP?"

- In the multidimensional model, data are organized into multiple dimensions (e.g. time, item, location, supplier), and each dimension contains multiple levels of abstraction defined by concept hierarchies. This organization provides users with the flexibility to view data from different perspectives.
- A number of OLAP data cube operations exist to materialize these different views, allowing interactive query and analysis of the data at hand. Hence, OLAP provides a user-friendly environment for interactive data analysis.
- Operations include:
  - Roll up (drill-up),
  - Drill down (roll down),
  - Slice and dice,
  - Pivot (rotate),
  - and more

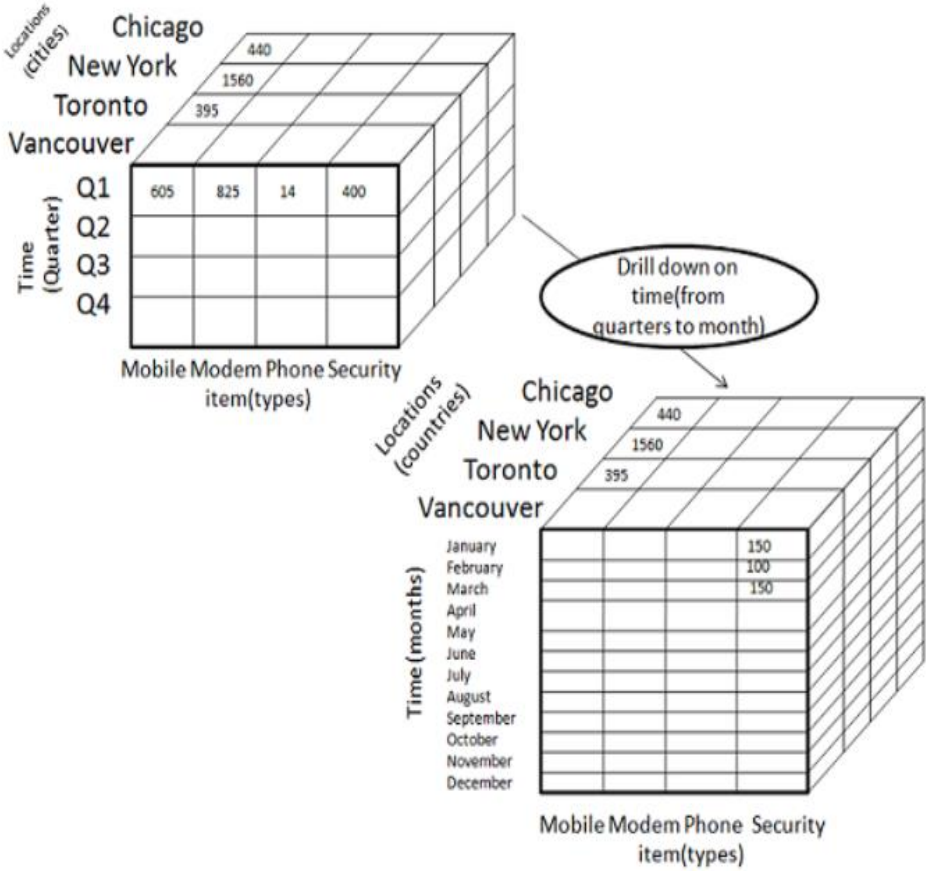
Typical OLAP Operations – Roll-up

- Roll-up performs aggregation on a data cube
  - By climbing up a concept hierarchy for a dimension
  - Or By dimension reduction
- Example: Roll-up is performed by climbing up a concept hierarchy for the dimension *location*.
- Initially, the concept hierarchy was defined as the total order "street < city < state < country".
- The data is grouped into cities rather than countries.
- On rolling up, the data is aggregated by ascending the *location* hierarchy from the level of city to the level of country.
- When roll-up is performed, one or more dimensions from the data cube can be removed.



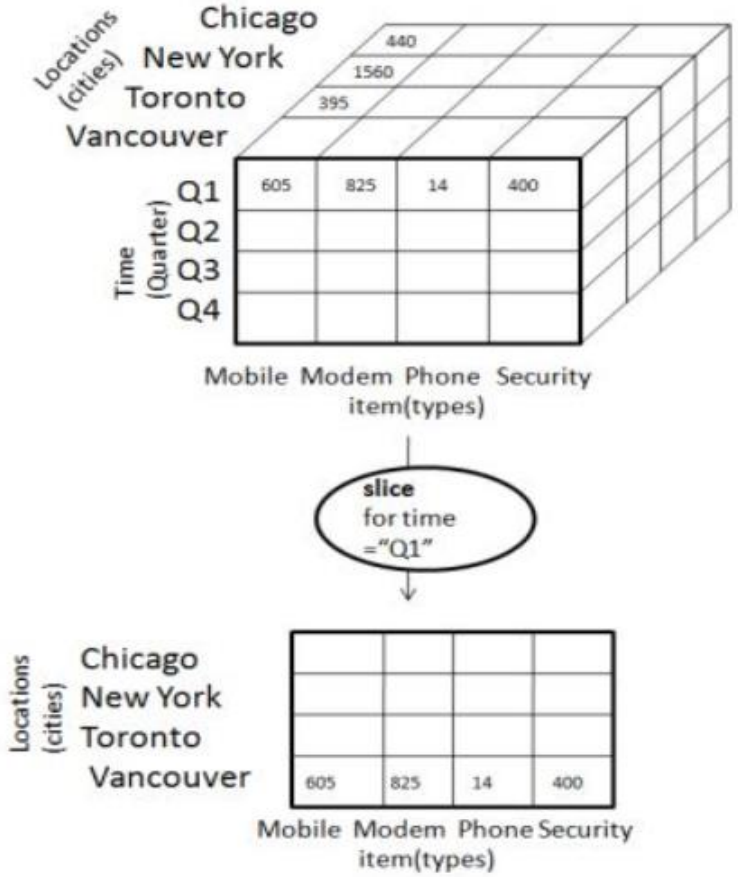
Typical OLAP Operations – Drill-down

- Drill-down is the reverse operation of roll-up. It is performed by
  - By stepping down a concept hierarchy for a dimension
  - Or By introducing a new dimension.
- Example: Drill-down is performed by stepping down a concept hierarchy for the dimension *time*.
- Initially, the concept hierarchy was defined as "day < month < quarter < year."
- On drilling down, the time dimension is descended from the level of quarter to the level of month.
- When drill-down is performed, one or more dimensions from the data cube are added.
- It navigates the data from less detailed data to highly detailed data.



Typical OLAP Operations – Slice

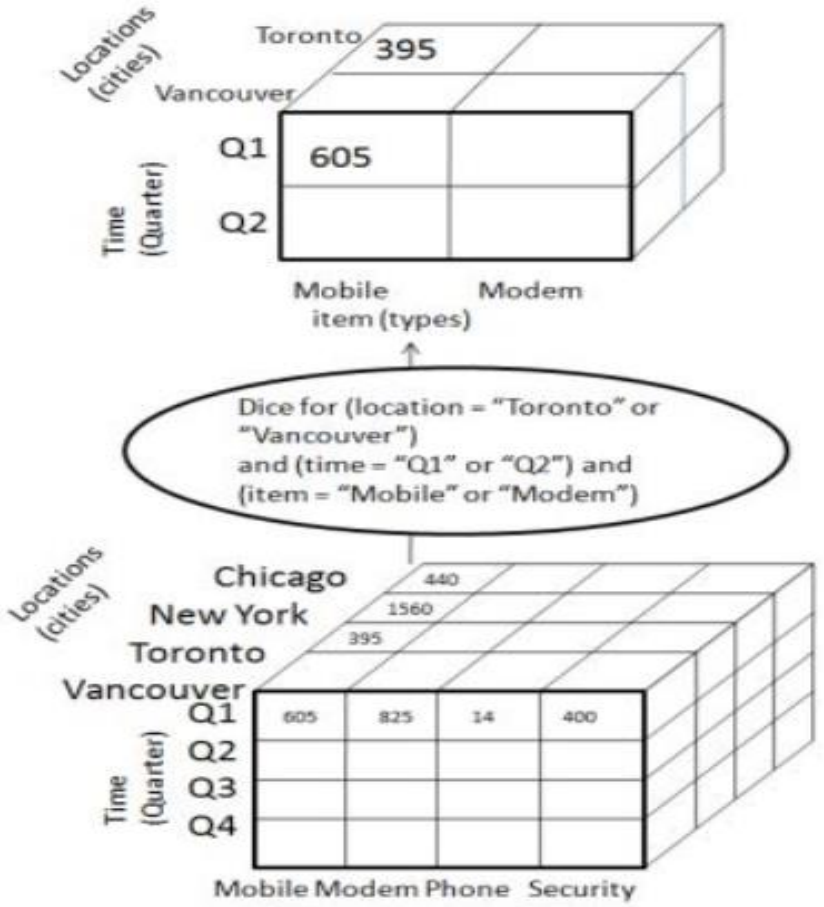
- The slice operation selects one particular dimension from a given cube and provides **a new sub-cube**.
- Here Slice is performed for the dimension *time* using the criterion time = "Q1".
- It will form a new sub-cube by selecting one dimension.



Typical OLAP Operations – Dice

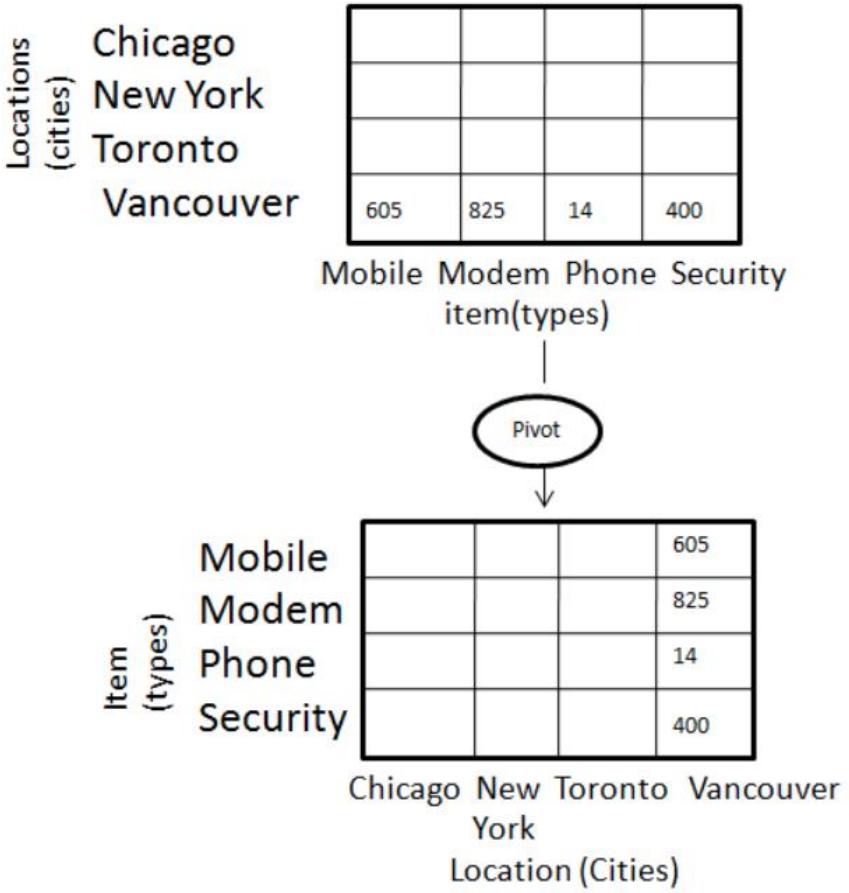
- Dice selects two or more dimensions from a given cube and provides **a new sub-cube**.

- Example: The dice operation on the cube based on the following selection criteria involves three dimensions.
  - (location = "Toronto" or "Vancouver")
  - (time = "Q1" or "Q2")
  - (item =" Mobile" or "Modem")



Typical OLAP Operations – Pivot

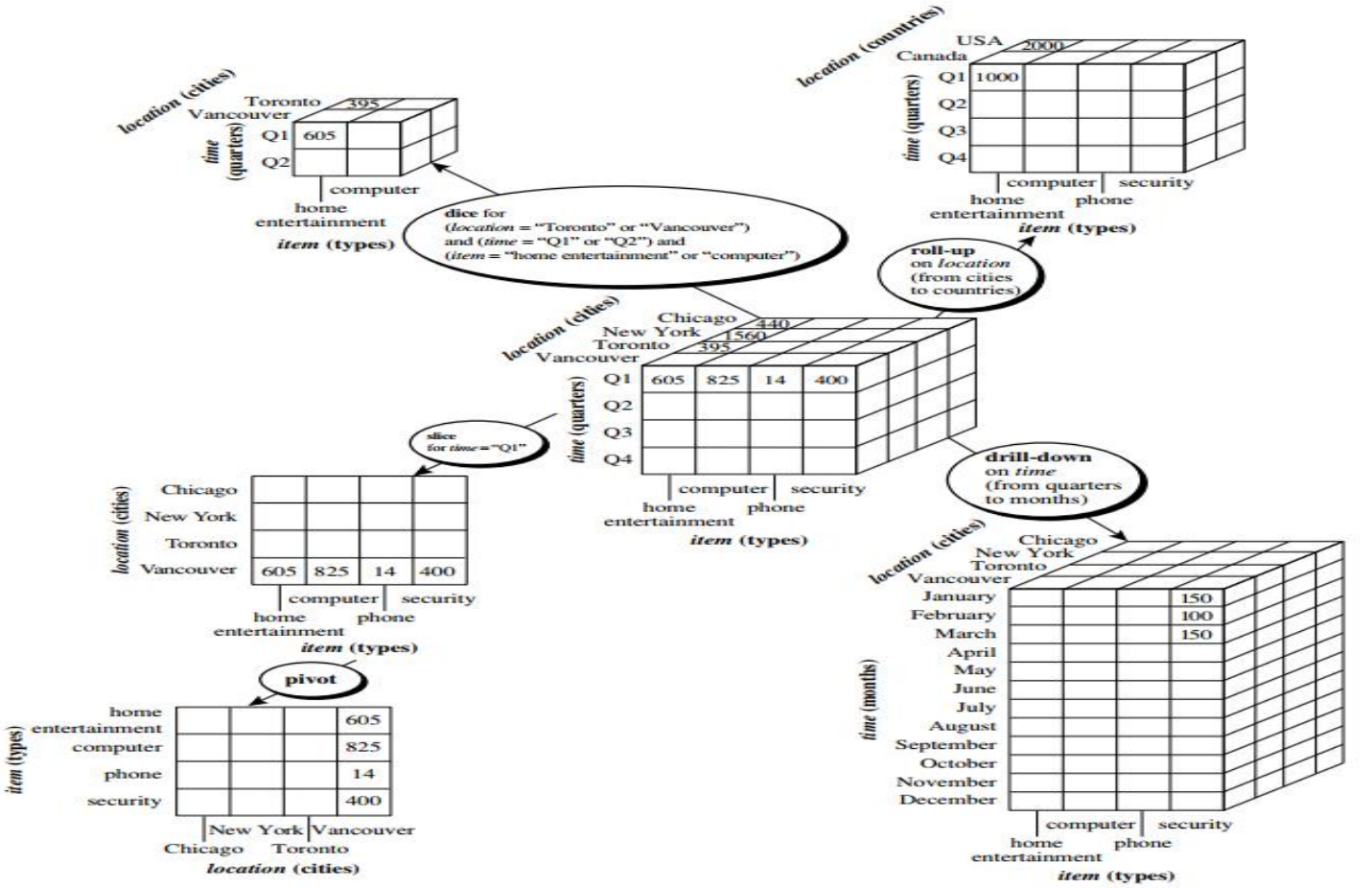
- The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data.



Typical OLAP Operations

- Roll up (drill-up): summarize data
  - *by climbing up hierarchy or by dimension reduction*
- Drill-down (roll-down): reverse of roll-up
  - *from higher level summary to lower level summary or detailed data, or introducing new dimensions*
- Slice and dice: *project and select*
- Pivot (rotate):
  - *reorient the cube, visualization, transform a 3D cube to a series of 2D planes*
- Other operations
  - *drill across: involving (across) more than one fact table*
  - *drill through: through the bottom level of the cube to its back-end relational tables (using SQL)*

Examples of Typical OLAP Operatiobs on Multidimensional Data



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Efficient Data Cube Computation

- Data cube can be viewed as a lattice of cuboids
  - The bottom-most cuboid is the base cuboid
  - The top-most cuboid (apex) contains only one cell
  - How many cuboids in an n-dimensional cube with L levels?

$$T = \prod_{i=1}^{n} (L_i + 1)$$

where L_i is the number of levels associated with dimension i. One is added to include the virtual top level, *all*.

  - If the cube has 10 dimensions, each of which has 5 levels (including *all,* the virtual top level), total number of cuboids can be $5^{10} \approx 9.8 \times 10^6$
- Materialization of data cube
  - Materialize every (cuboid) (**full materialization**), none (**no materialization**), or some (**partial materialization**)
  - Selection of which cuboids to materialize
    - Based on size, sharing, access frequency, etc.

Indexing OLAP Data: Bitmap Index

To facilitate efficient data accessing, most data warehouse systems support index structures and materialized views (using cuboids)

- Index on a particular column
- Each value in the column has a bit vector: bit-op is fast
- The length of the bit vector: # of records in the base table
- The *i*-th bit is set to 1, if the *i*-th row of the base table has the value for the indexed column

Base table

Cust	Region	Type
C1	Asia	Retail
C2	Europe	Dealer
C3	Asia	Dealer
C4	America	Retail
C5	Europe	Dealer

Index on Region

RecID	Asia	Europe	America
1	1	0	0
2	0	1	0
3	1	0	0
4	0	0	1
5	0	1	0

Index on Type

RecID	Retail	Dealer
1	1	0
2	0	1
3	0	1
4	1	0
5	0	1

OLAP Server Architectures

- Relational OLAP (ROLAP)
  - Use relational or extended-relational DBMS to store and manage warehouse data and OLAP middle ware
  - Include optimization of DBMS backend, implementation of aggregation navigation logic, and additional tools and services
  - Greater scalability
- Multidimensional OLAP (MOLAP)
  - Sparse array-based multidimensional storage engine
  - Fast indexing to pre-computed summarized data
- Hybrid OLAP (HOLAP) (e.g., Microsoft SQLServer)
  - Flexibility, e.g., low level: relational, high-level: array

ROLAP VS MOLAP

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	MOLAP	ROLAP
1	Information retrieval is fast.	Information retrieval is comparatively slow.
2	Uses sparse array to store datasets.	Uses relational table.
3	Easy to use for beginners	Comparatively difficult to use for beginners
4	Maintains a separate database for data cubes.	It may not require space other than available in the Data warehouse.

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Attribute-Oriented Induction

- The attribute-oriented induction (AOI) approach on concept description was proposed in 1989 (KDD '89 workshop)
- Not confined to categorical data nor particular measures
- How is it done?
  - Collect the task-relevant data (*initial relation*) using a relational database query
  - Perform generalization by attribute removal or attribute generalization
  - Apply aggregation by merging identical, generalized tuples and accumulating their respective counts
  - Interact with users for knowledge presentation by mapping generalized relation into different forms (e.g. charts or rules)

Basic Principles of Attribute-Oriented Induction

- Data focusing: task-relevant data, including dimensions, and the result is the *initial relation*
- Attribute-removal: remove attribute *A* if there is a large set of distinct values for *A* but (1) there is no generalization operator on *A*, or (2) *A*'s higher-level concepts are expressed in terms of other attributes
- Attribute-generalization: If there is a large set of distinct values for *A*, and there exists a set of generalization operators on *A*, then select an operator and generalize *A*
- Attribute-threshold control: typical 2-8, specified/default
- Generalized relation threshold control: control the final relation/rule size

Attribute-Oriented Induction: An Example

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Example: Describe the general characteristics of graduate students in the University database

- Step 1. Fetch relevant set of data using an SQL statement, e.g.,

  Select * (i.e., name, gender, major, birth_place, birth_date, residence, phone#, gpa)

  from student

  where student_status in {“Msc”, “MBA”, “PhD” }

- Step 2. Perform attribute-oriented induction
- Step 3. Present results in generalized relation, cross-tab, or rule forms

Class Characterization: An Example

Initial Relation

Name	Gender	Major	Birth-Place	Birth_date	Residence	Phone #	GPA
Jim Woodman	M	CS	Vancouver,BC, Canada	8-12-76	3511 Main St., Richmond	687-4598	3.67
Scott Lachance	M	CS	Montreal, Que, Canada	28-7-75	345 1st Ave., Richmond	253-9106	3.70
Laura Lee	F	Physics	Seattle, WA, USA	25-8-70	125 Austin Ave., Burnaby	420-5232	3.83
…	…	…	…	…	…	…	…
Removed	Retained	Sci,Eng, Bus	Country	Age range	City	Removed	Excl, VG,..

Prime Generalized Relation

Gender	Major	Birth_region	Age_range	Residence	GPA	Count
M	Science	Canada	20-25	Richmond	Very-good	16
F	Science	Foreign	25-30	Burnaby	Excellent	22
…	…	…	…	…	…	…

Gender \ Birth_Region	Canada	Foreign	Total
M	16	14	30
F	10	22	32
Total	26	36	62

Summary

- Data warehousing: A multi-dimensional model of a data warehouse
  - A data cube consists of *dimensions & measures*
  - Star schema, snowflake schema, fact constellations
  - OLAP operations: drilling, rolling, slicing, dicing and pivoting
- Data Warehouse Architecture
  - Multi-tiered architecture
- Implementation: Efficient computation of data cubes
  - Indexing OLAP data: Bitmap index and join index
  - OLAP servers: ROLAP, MOLAP, HOLAP
- Data generalization: Attribute-oriented induction

Have a question?

